Groups and Lie Algebras: Exercise Sheet 3

Exercise 3.1: Some properties of characters

Let V and W be a finite dimensional representation of a finite group G with characters χ_V and χ_W. Compute the character of the representations

$$V \oplus W, \quad V \otimes W, \quad V^*, \quad \mathrm{Hom}(V, W), \quad \Lambda^2 V, \quad \mathrm{Sym}^2 V.$$

Answer:

To compute the characters, we use the definitions of the above representations and the properties of the trace:

- The direct sum $V \oplus W$ is defined as the vector space $V \oplus W = \{(v, w) \mid v \in V, w \in W\}$ with the action of G given by $g \cdot (v, w) = (g \cdot v, g \cdot w)$. The trace of this representation is the sum of the traces of the individual representations, so we have $\chi_{V \oplus W}(g) = \chi_V(g) + \chi_W(g)$.
- The tensor product $V \otimes W$ is defined as the vector space generated by the elements $v \otimes w$ with $v \in V$ and $w \in W$, with the action of G given by $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$. The trace of this representation is the product of the traces of the individual representations, so we have $\chi_{V \otimes W}(g) = \chi_V(g) \cdot \chi_W(g)$.
- The dual representation V^* is defined as the vector space of linear functionals on V, with the action of G given by $(g \cdot f)(v) = f(g^{-1} \cdot v)$ for $f \in V^*$ and $v \in V$. The trace of this representation is the complex conjugate of the trace of the original representation, so we have $\chi_{V^*}(g) = \overline{\chi_V(g)}$.
- The verctor space of homomorphisms $\mathrm{Hom}(V, W)$ is the set of maps from V to W that preserve the group structure. The action of G is given by $(g \cdot \phi)(v) = g_W \cdot \phi(g_V^{-1} \cdot v)$. To compute the character we first need to show $\mathrm{Hom}(V, W) \cong V^* \otimes W$. To show this we define the map

  $$\Phi : V^* \otimes W \to \mathrm{Hom}(V, W), \quad (\phi \otimes w) \mapsto (v \mapsto \phi[v] \cdot w)$$

  This is indeed an isomorphism since

  $$\dim(V^* \otimes W) = \dim(V) \cdot \dim(W) = \dim(\mathrm{Hom}(V, W))$$

To show that Φ is injective, we evaluate the zero map on the basis vectors e_k of V:

$$\begin{aligned}\Phi\left(\sum_{i,j} a_{ij}\,\phi_i \otimes w_j\right)[e_k] &= 0\\ \sum_{i,j} a_{ij}\,\phi_i[e_k]\cdot w_j &= 0\\ \sum_{i,j} a_{ij}\,\delta_{ik}\cdot w_j &= 0\\ \sum_{j} a_{kj}\,w_j &= 0\end{aligned}$$

Since the w_j are linearly independent, we have $a_{kj} = 0$ for all j. Since this is true for all k, we have $a_{ij} = 0$ for all i, j and thus Φ is injective. Therefore we have $\mathrm{Hom}(V, W) \cong V^* \otimes W$ and we can compute the character as $\chi_{\mathrm{Hom}(V,W)}(g) = \chi_{V^*\otimes W}(g) = \chi_{V^*}(g)\cdot\chi_W(g) = \overline{\chi_V(g)}\cdot\chi_W(g)$.

- To compute the characters of the symmetric and antisymmetric rank 2 tensors, we notice that

$$V \otimes V = \Lambda^2 V \oplus \mathrm{Sym}^2 V$$

This can be seen by simply writing down the definitions. For all $v, w \in V$ we have

$$v \otimes w = \frac{1}{2}(v \otimes w + w \otimes v) + \frac{1}{2}(v \otimes w - w \otimes v) = v \vee w + v \wedge w$$

Therefore we can compute the characters of the symmetric and antisymmetric rank 2 tensors as

$$\chi_{V\otimes V}(g) = \chi_{\Lambda^2 V}(g) + \chi_{\mathrm{Sym}^2 V}(g)$$

Futhermore consicer the eigenvalues of the representation V for a fixed $g \in G$. Let these eigenvalues be $\lambda_1, \ldots, \lambda_n$. Then we have

$$\chi_{V\otimes V}(g) = \sum_{i,j} \lambda_i\lambda_j$$

Futhermore, if λ_i are the eigenvalues of g in V, then the eigenvalues of g^2 in V are λ_i^2. Therefore we have

$$\chi_V(g^2) = \sum_i \lambda_i^2$$

We can now compute the characters of the symmetric and antisymmetric rank 2 tensors as

$$\chi_{\Lambda^2 V}(g) = \sum_{i<j} \lambda_i\lambda_j = \frac{1}{2}\left(\sum_{i,j}\lambda_i\lambda_j - \sum_i \lambda_i^2\right) = \frac{1}{2}(\chi_{V\otimes V}(g) - \chi_V(g^2))$$

$$\chi_{\mathrm{Sym}^2 V}(g) = \sum_{i\leq j} \lambda_i\lambda_j = \frac{1}{2}\left(\sum_{i,j}\lambda_i\lambda_j + \sum_i \lambda_i^2\right) = \frac{1}{2}(\chi_{V\otimes V}(g) + \chi_V(g^2))$$

■

Exercise 3.2: Quaternions

The quaternion group consists of 8 elements

$$\pm 1, \quad \pm i, \quad \pm j, \quad \pm k.$$

The defining relations are

$$i^2 = j^2 = k^2 = -1, \qquad i = jk = -kj, \qquad j = ki = -ik, \qquad k = ij = -ji.$$

a) First find 1 four-dimensional representations of Q_8.

Answer:

We start by writing the elements of Q_8 as the vectors

$$1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad i = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad j = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad k = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

We can then define a representation by left multiplication. To find the explicit form we use the above relations. 1 acts as the identity, so we start by considering the action of i:

$$\rho(i) \cdot 1 = 1, \quad \rho(i) \cdot i = 1, \quad \rho(i) \cdot j = k, \quad \rho(i) \cdot k = -j$$

so we have

$$\rho(i) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

We can do the same for j and k:

$$\rho(j) = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad \rho(k) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

To check if this representation is irreducible, we compute the inner product of the character with itself:

$$\langle \chi, \chi \rangle = \frac{1}{|Q_8|} \sum_{g \in Q_8} |\chi(g)|^2 = \frac{1}{8}(2 \cdot 4^2 + 6 \cdot 0^2) \neq 1$$

This means this representation is not irreducible. ■

b) Find 4 one-dimensinal representations of Q_8. Are they irreducible?

Answer:

To find the one-dimensional representations, we find a homomorphism from Q_8 to $\mathbb{C}^\times$. We start by considering $\rho(1) = 1$ and $\rho(-1)^2 = \rho(1) = 1$. This means $\rho(-1) \in \{\pm 1\}$. Similarly, we have $\rho(i)^2 = \rho(j)^2 = \rho(k)^2 = \rho(-1)$. Thus if $\rho(-1) = 1$ then we have $\rho(i) = \rho(j) = \rho(k) \in \{\pm 1\}$ and if $\rho(-1) = -1$ then we have $\rho(i), \rho(j), \rho(k) \in \{\pm i\}$. From the relations above we can exclude $\rho(-1) = -1$ since $\rho(i) = \rho(j)\rho(k) = \pm i^2 \notin \{\pm i\}$. Thus we can write down all one-dimensional representations by choosing $\rho(i), \rho(j), \rho(k) \in \{\pm 1\}$. But since $\rho(k) = \rho(i)\rho(j)$, only four choices are possible. We can write them explicitly in the following table:

	1	-1	i	j	k
ρ_1	1	1	1	1	1
ρ_2	1	1	-1	-1	1
ρ_3	1	1	-1	1	-1
ρ_4	1	1	1	-1	-1

They are all irreducible since they are one-dimensional. ■

c) There is also a 2-dimensional irreducible representation. To find it, think of quantum mechanics.

Answer:

To find the 2 dimensional representation, we use the pauli matrices. The pauli-matrices are unitary and self-adjoint, therefore we multiply them by i to get a representation of Q_8:

$$\rho(i) = i\sigma_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \rho(j) = i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \rho(k) = i\sigma_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

Again we check if this representation is irreducible by computing the inner product of the character with itself:

$$\langle \chi, \chi \rangle = \frac{1}{|Q_8|} \sum_{g \in Q_8} |\chi(g)|^2 = \frac{1}{8}(2 \cdot 2^2 + 6 \cdot 0^2) = 1$$

This means this representation is irreducible. ■

d) Are these all irreducible representations? Explain.

Answer:

To check if wa have all irreducible representations, we use the following two facts:

- $|\text{ Conj(G) }| = \sum_{\rho \text{ irrep.}} 1$
- $|G| = \sum_{\rho \text{ irrep.}} (\dim \rho)^2$

First we compute the conjugacy classes of Q_8:

$$\text{Conj}(1) = \{g \in Q_8 \mid g = q1q^{-1}\} = \{1\}$$

$$\text{Conj}(-1) = \{g \in Q_8 \mid g = q(-1)q^{-1}\} = \{-1\}$$

Are trivialy given. From the defining relations we can also compute the conjugacy

classes of i, j and k:

$$i = jk \quad j = ki \implies -j = -ki = -kjk = kjk^{-1}$$

Similarly we can compute all conjugations of i, j and k and we find that

$$\mathrm{Conj}(i) = \{i, -i\}, \quad \mathrm{Conj}(j) = \{j, -j\}, \quad \mathrm{Conj}(k) = \{k, -k\}$$

So we have 5 conjugacy classes, which means we need 5 irreducible representations. Therefore the 4 one-dimensional representations and the 2-dimensional representation we found are all irreducible representations of Q_8.

We can also derive the fact that the irreducible representations of Q_8 are four 1-dimensional and one 2-dimensional representation using the second fact above. We have

$$|Q_8| = 8 = 2 \cdot 2^2 = 2^2 + 4 \cdot 1^2 = 8 \cdot 1^2$$

Since we need five irreducible representations, we can only have four 1-dimensional and one 2-dimensional representation. ■

e) Compute the character table.

Answer:

We can now compute the character table of Q_8 using the representations we found and the conjugacy classes:

	$\{1\}$	$\{-1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
ρ_1	1	1	1	1	1
ρ_2	1	1	-1	-1	1
ρ_3	1	1	-1	1	-1
ρ_4	1	1	1	-1	-1
ρ_5	2	-2	0	0	0

■

Exercise 3.3: The symmetric group S_4 and its representations

a) Determine the conjugacy classes of S_4 and associate Young diagrams and conjugacy classes.

Answer:

■

b) There are two 1-dimensional representations of S_4. What are they?

Answer:

■

c) Consider the 4-dimensional standard representation of S_4 on $\mathbb{R}^4$. Decompose it into a

3-dimensional irreducible representation and a 1-dimensional representation.

Answer:

d) Take the tensor product of the 3-dimensional representation of the previous part and the alternating representation. It is again irreducible. How many other irreducible representations are there? What are their dimensions?

Answer:

e) Write down the character table.

Answer:

Exercise 3.4: Particle in a box

The Hilbert space of a particle in a box of length l is $L^2([-l,l]^2)$ with the boundary conditions

$$\psi|_{\partial[-l,l]^2} = 0$$

and Hamiltonian

$$\mathbb{H} = -\frac{\hbar^2}{2m}\left(\partial_x^2 + \partial_y^2\right).$$

Show that this system has D_4 symmetry. Solve the Schrödinger equation and find an energy level whose eigenfunctions transform in the 2-dimensional representation of D_4. What is the lowest energy level that does not transform in an irreducible representation of D_4.

Answer: